MCQsim25

Software Documentation

Input Component: Geometry GUIs

Overview

MCQsim25 is a fault-based seismic simulation code. Its workflow is organized into three main stages:

Stage	Description
1. Input	Fault geometry definition and loading assignment via MATLAB GUIs
2. Main Code	Stochastic earthquake simulation engine
3. Output	Result visualization and analysis via MATLAB GUIs

This document covers Stage 1 — the Input GUIs. Two alternative Geometry GUIs are available on GitHub. The user selects one depending on whether the fault mesh already exists (as an STL file) or needs to be constructed from scratch. Both GUIs produce a data file consumed by the next step in the pipeline.

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1 Input Component: Geometry GUIs

The Input Component consists of two Geometry GUIs. They serve the same purpose — defining fault geometry and assigning tectonic loading conditions — but differ in how the fault mesh is obtained:

GUI	Use case
A_GEOMfromSTL	Fault mesh already exists as one or more STL files
A_GEOMmanual	Fault geometry must be built interactively from rectangular segments

After completing either GUI the user saves a model file. This file is the direct input to the B_RoughStrength, which assigns fault roughness and friction parameters before the main simulation code is run.

General rule: follow the numbered buttons in the GUI (1, 2, 3, …) to progress through the workflow in the correct order.

2 A_GEOMfromSTL — Import Existing Mesh

Use this GUI when the fault surface has already been meshed and each fault segment is stored as a separate STL file. The GUI does not alter the mesh geometry; its sole purpose is to assign tectonic loading conditions and export the data in the format required by MCQsim25.

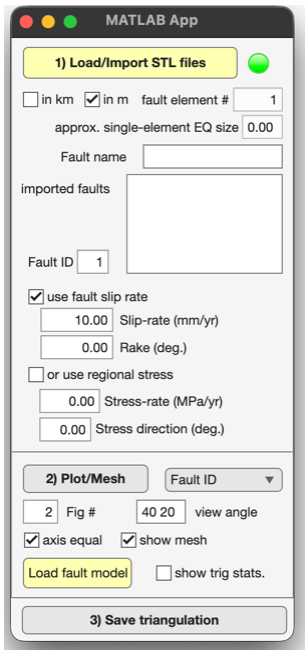

Figure 1: *The A_GEOMfromSTL GUI. Three numbered sections guide the workflow: (1) Load/Import STL files, (2) Plot/Mesh for visualization, and (3) Save triangulation. The green indicator lamp confirms a successful import.*

The GUI layout directly reflects the workflow: work top-to-bottom through the numbered controls. Once an STL file is loaded successfully, the status lamp turns green and summary information is displayed. Loading options, fault naming, and loading-condition fields are all visible in a single panel.

2.1 Input Requirements

STL File Format

Each fault segment must be supplied as a separate STL file in ASCII format:

```
solid <n>
  facet normal <nx> <ny> <nz>
    outer loop
      vertex <x1> <y1> <z1>
      vertex <x2> <y2> <z2>
      vertex <x3> <y3> <z3>
    endloop
  endfacet
  ...
endsolid <n>
```

Coordinate System Conventions

Requirement	Details
Units	Metric only (metres or kilometres). Decimal degrees are not accepted.
Depth sign convention	Depth must be negative (below the surface).
Coordinate system	All STL files must share a common coordinate system.
Unit selection	Correctly declare metres or kilometres when loading — see the in km / in m checkboxes in the GUI.

2.2 Step-by-Step Workflow

Step 1 — Load / Import STL Files

Press button 1) Load/Import STL files to open a file browser and select one or more STL files. On successful import:

- The status indicator lamp turns green.
- The number of imported fault elements is reported in the fault element # field.
- Fault names are derived automatically from the STL file names and appear in the imported faults list box.
- The approximate single-element EQ size is computed from the average element area together with standard earthquake scaling relationships, estimating the magnitude of a single-element rupture event.

Note: *If the lamp does not turn green, verify that the STL files are in ASCII format, that the correct unit (m or km) is selected, and that depth values are negative.*

Step 2 — Edit Fault Names and IDs

After importing, each fault name can be edited in the Fault name field. Changes are reflected immediately in the imported faults list box.

Fault ID Each fault section carries a numeric Fault ID. Sections sharing the same ID are treated as belonging to the same fault. This grouping governs how fault roughness is applied: roughness is allowed to extend across sections with the same ID, producing a continuous rough surface across multiple segments. Change the ID field to merge or separate sections.

Step 3 — Assign Loading Conditions

An average loading rate is assigned to each fault section. Two loading types are available:

Option A: Prescribed Fault Slip Rate

Tick use fault slip rate and fill in:

Field (GUI label)	Description
Slip-rate (mm/yr)	Must be a positive value.
Rake (deg.)	Slip direction relative to the fault strike. See convention table below.

Rake value	Fault motion
0°	Left-lateral (sinistral) strike-slip
90°	Reverse (thrust)
180° / −180°	Right-lateral (dextral) strike-slip
−90°	Normal

Note: *The local coordinate system is: x-axis along strike, y-axis up-dip, z-axis fault-normal.*

Option B: Regional Stress Rate

Tick or use regional stress and fill in:

Field (GUI label)	Description
Stress-rate (MPa/yr)	Magnitude of the stress rate. Negative values indicate compression.
Stress direction (deg.)	Azimuthal trend of the stress axis (degrees from North).

In the main code, the assigned average loading rate is spatially redistributed across fault elements to minimise numerical artefacts from the back-slip method.

Step 4 — Visualise the Mesh

Use the 2) Plot/Mesh section to display the fault mesh. Select a colouring mode (e.g. Fault ID) from the drop-down, set a figure number and view angle, then click Plot/Mesh. Verify the geometry and loading before proceeding.

Step 5 — Save and Continue

Press 3) Save triangulation to write the model file. This output is the direct input to the B_RoughStrength GUI.

3 A_GEOMmanual — Build Mesh from Rectangles

Use this GUI when no pre-existing STL mesh is available. Faults are constructed interactively from rectangular segments, which the GUI then triangulates into a mesh. The same fault-naming, Fault ID, and loading conventions described in Section 2 apply here as well.

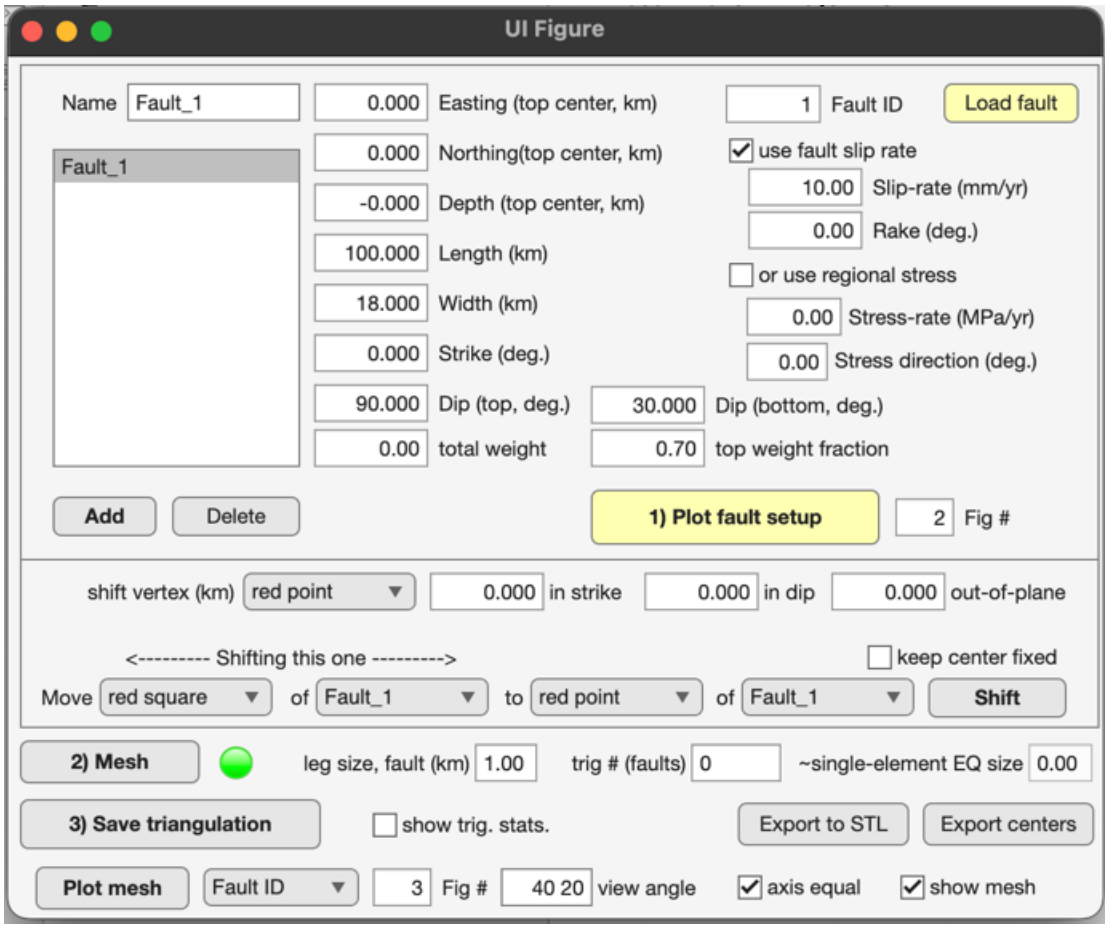

Figure 2: The A_GEOMmanual GUI. The upper half contains all geometric parameters for defining and editing fault segments (name, list box, Add/Delete, geometry fields, and loading assignment). The lower half provides vertex-shifting controls, segment-connection tools, and the 2) Mesh and 3) Save triangulation buttons.

3.1 Fault Definition

Geometric Parameters

Each fault is initially defined as a rectangle. Specify the following parameters in the GUI:

Parameter (GUI label)	Description
Easting (top center, km)	Horizontal easting of the fault top-centre point.
Northing (top center, km)	Horizontal northing of the fault top-centre point.
Depth (top center, km)	Depth of the top-centre point. Must be negative.
Length (km)	Along-strike fault length.
Width (km)	Down-dip fault width.
Strike (deg.)	Strike angle in degrees from North.
Dip (top, deg.)	Dip angle at the top edge of the fault.

Parameter (GUI label)	Description
Dip (bottom, deg.)	Dip angle at the bottom edge of the fault.
total weight	Bézier handle length as a fraction of fault width. Set to 0 for a planar fault.
top weight fraction	Fraction of total weight allocated to the upper Bézier handle.

Two dip values are accepted to allow listric fault geometries. For a uniform planar dip, set both values to the same angle.

Listric Faults

Listric fault shapes are implemented using cubic Bézier curves. In a cubic Bézier curve (Figure 3), the path between a start and end point is shaped by two control handles whose positions determine the curvature. In MCQsim25, the fault profile from top to bottom follows this curve, with the two handles controlling how the dip varies with depth.

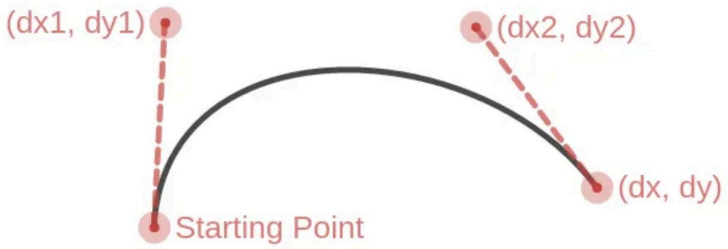

Figure 3: *Illustration of a cubic Bézier curve. The Starting Point and endpoint (dx, dy) correspond to the fault top and bottom edges. The two control handles (dx1, dy1) and (dx2, dy2) shape the curvature of the fault profile. In MCQsim25, the combined length of these handles relative to the fault width is the total weight parameter, and top weight fraction controls how that length is split between the upper and lower handle. (Source: medium.com/@bragg/cubic-bezier-curves-with-svg-paths)*

A non-zero total weight combined with different top and bottom dip values produces a curved (listric) fault surface. Set total weight to 0 for a planar fault. Both shallowing-upward and steepening-upward geometries are possible, enabling the construction of flower structures.

> ***Note:*** *Experiment with different combinations of total weight, top weight fraction, and top/bottom dip values to achieve the desired shape.*

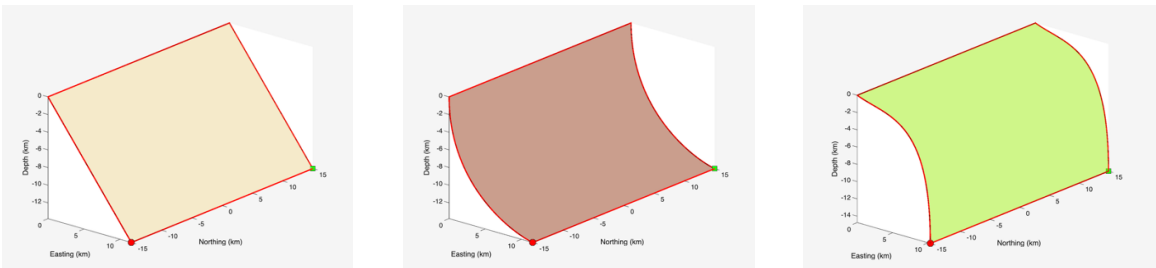

Figure 3: *Example for creating a listric fault from a dipping fault.Left: the initial geometry with a total weight of zero. Middle: Total weight is one, with a top fraction of 0.7 and a top dip of 90deg and bottom dip of 10deg. Right: Flipped setup, where the top dip is 20deg and bottom dip is 90deg, creating part of a flower structure.*

3.2 Step-by-Step Workflow

Step 1 — Define and Preview the Fault

Enter the geometric parameters for the first fault segment and press 1) Plot fault setup to preview the rectangular outline before meshing. The currently selected fault is always highlighted with a red outline.

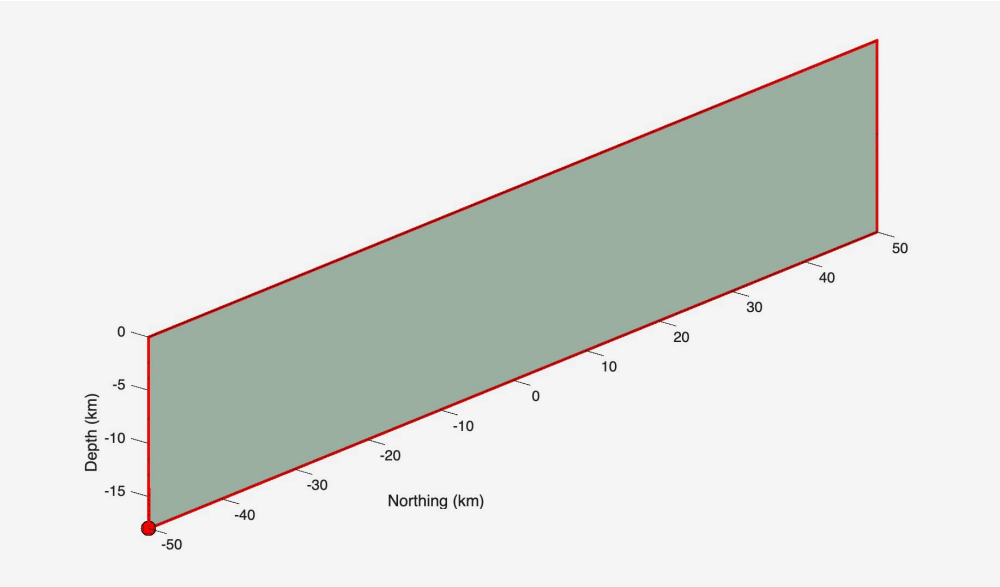

Figure 5: A single fault segment displayed as a rectangle after pressing 1) Plot fault setup. The red outline indicates the currently selected fault. This example shows a fault of 100 km length and 18 km width with 90° dip, prior to meshing.

Step 2 — Manage Fault Segments

Adding a fault: Click Add to create a new segment with a random name and default values.

Editing a fault: Select the segment in the Name list box. The parameter fields update to show that segment. Modify values and press 1) to preview.

Deleting a fault: Select the segment and click Delete.

Fault names and IDs: Edit the Name field and Fault ID field using the same conventions described in Section 2.2, Step 2.

Step 3 — Fine-tune Vertices

Individual fault corners can be repositioned to adjust the geometry or connect adjacent segments. In the shift vertex section, select the target fault from the list box and choose one of the four coloured vertices from the drop-down:

Vertex label	Position on fault rectangle
red point	Top-left corner
blue point	Top-right corner
green point	Bottom-left corner
yellow point	Bottom-right corner

Specify the displacement in three directions (all in km):

Field	Description
in strike	Displacement along the fault strike direction.

Field	Description
in dip	Displacement along the fault dip direction.
out-of-plane	Displacement perpendicular to the fault plane.

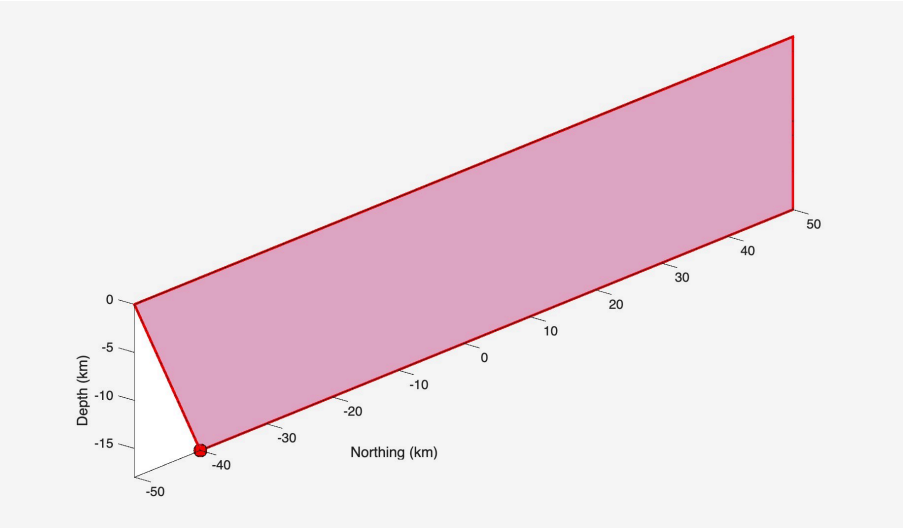

Figure 6: *A single fault after shifting the bottom-left vertex (red point) by 10 km in the strike direction. The filled red circle marks the selected vertex. Compare with Figure 5 to see the effect of the shift.*

keep center fixed: When NOT ticked, shifting a vertex translates the entire fault segment rigidly. When ticked, only the selected vertex moves, deforming the fault shape. This distinction becomes important when connecting two fault segments (see Step 4).

Step 4 — Connect Fault Segments

When working with multiple fault segments, the Connect faults panel allows a vertex of one segment to be snapped onto a vertex of another, creating a geometrically connected fault system. The workflow is illustrated in Figures 6– 9.

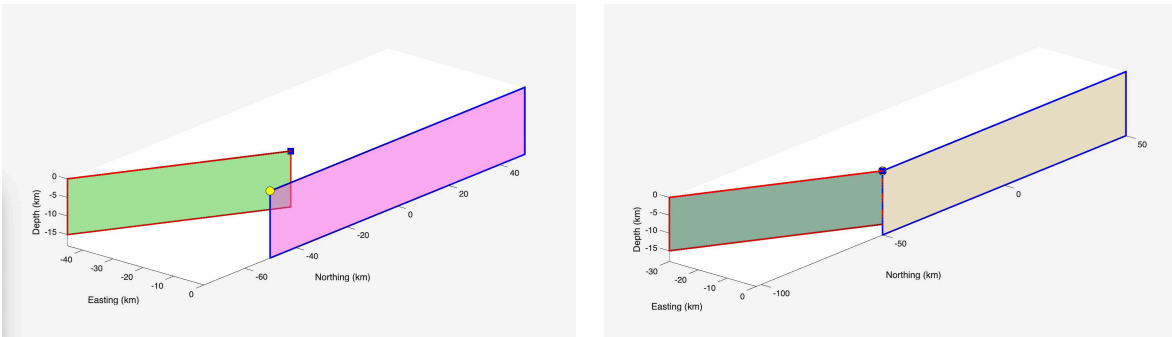

Figure 7: *Two separate fault segments before and after connection. Left: The segment outlined in red (left, green fill) is currently selected. The yellow circle marks the bottom-right vertex (yellow point) of Fault_1, which will be snapped to a vertex on Fault_2. The blue-outlined segment (pink fill) is Fault_2. Pressing the "Shift" button will then move the blue square, (and with it the other vertices as well), to the location of the yellow circle. Right: The new fault geometry after shifting. Note the mismatch at the bottom of the fault. This was intentiona as one fault as a width of 15 km and the other a width of 18 km. The following step allws to "close that gap".*

In the Shifting this one panel, specify the source vertex, source fault, target vertex, and target fault, then press Shift:

- Move [vertex] of [source fault] → to [vertex] of [target fault]

The effect of the keep center fixed option is shown in Figure 8. Press Shift to apply, then plot to verify the result before proceeding to meshing.

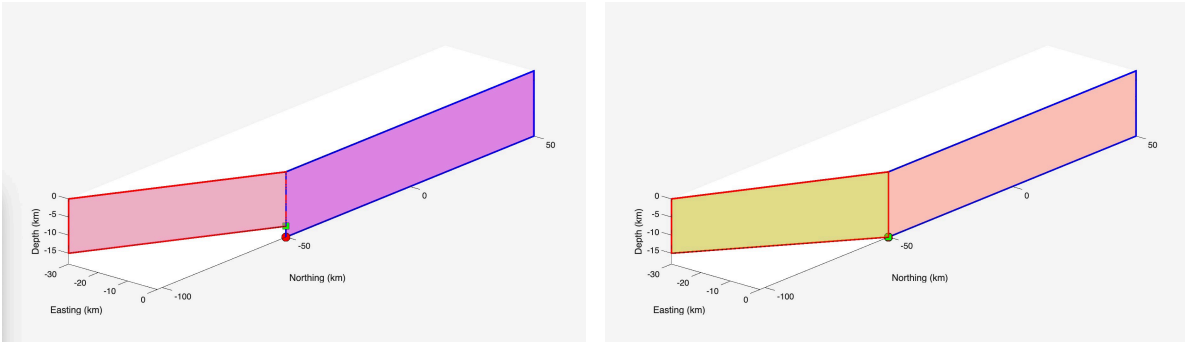

Figure 8: *Next, we can snap the lower green square to the red circle -here moving the upper vertex (green) downward to the depth of the larger fault. Left: Before shifting. Right: After shifting -while keeping the fault's position otherwise fixed.*

Step 5 — Set Mesh Resolution

Enter a value in the leg size, fault (km) field. The mesher builds equilateral triangles; this value is the target side length of each triangle. Smaller values produce finer meshes with more elements.

Step 6 — Generate the Mesh

Press 2) Mesh to triangulate all fault surfaces. The status lamp turns green when meshing completes successfully.

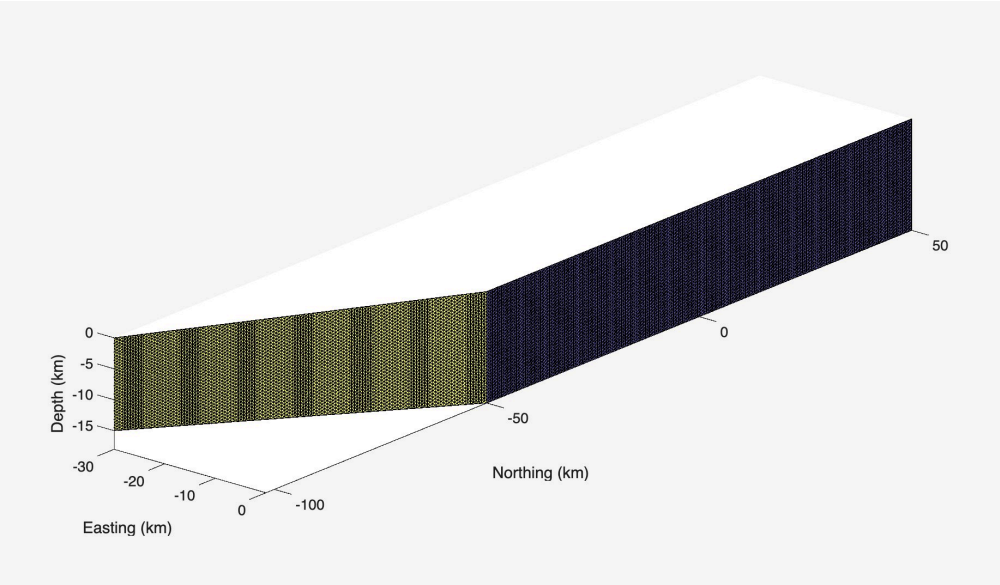

Figure 9: *The two connected fault segments after meshing, coloured by Fault ID (yellow-green for Fault_1, dark blue for Fault_2). The triangular element mesh is visible across both surfaces. Use Plot mesh with show mesh ticked to inspect mesh quality and element distribution. Note that the vertical bands observable in this figure are not "real" but merely a plotting artefact related to screen resolution.*

Step 7 — Assign Loading Conditions

Follow the same procedure as Section 2.2, Step 3 to assign either a slip rate or a stress rate to each fault section.

Step 8 — Save and Continue

Press 3) Save triangulation to write the model file. Optionally, use Export to STL to save the mesh as STL files for sharing or use in external software. The saved model file is then passed to the B_ GUI.

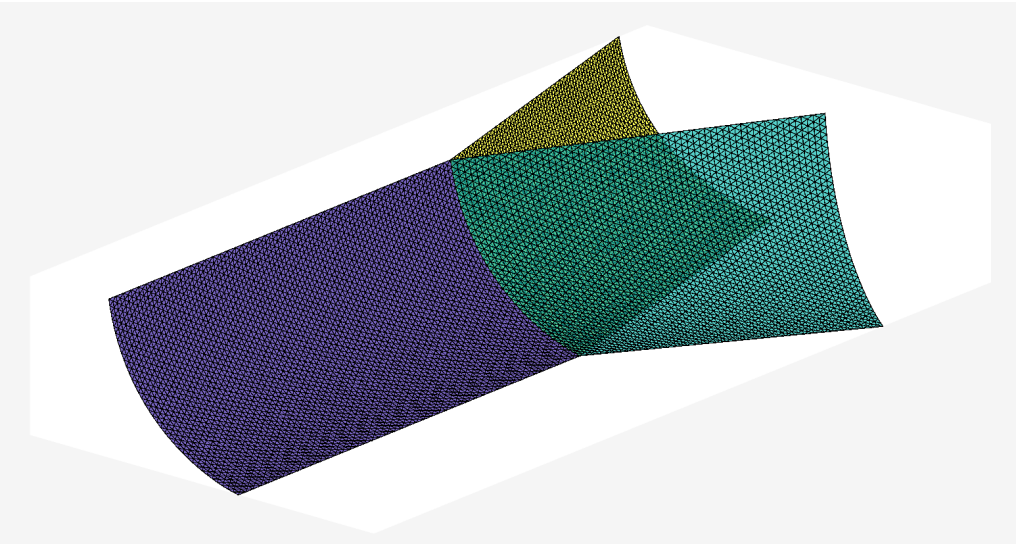

Figure 10: *The above-outlined concepts of joining faults and defining listric faults can be combined. Here an example of three listric faults forming a fork intersection.*

4 Next Steps

After completing either Geometry GUI and saving the model, the workflow continues:

Step	Description
B_RoughStrength GUI	Assign fault roughness and friction parameters to the saved geometry model.
Main Code	Run the stochastic earthquake simulation.
Output GUIs	Visualise and analyse simulation results.

Note: *Detailed documentation for the B_ GUI, the main simulation code, and the output GUIs will be provided in subsequent sections of this document.*

MCQsim25

Software Documentation

Input Component: B_RoughStrength GUI

Overview

The B_RoughStrength GUI is the second and final input step before running the main simulation code. It loads the triangulated fault model produced by either A_ GUI (stored as a .ftrg binary file) and assigns two classes of properties to the fault surface: geometric roughness and frictional strength. The outputs of this GUI — a .rgh file and a .stg file — are the direct inputs to the main simulation code.

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5 B_RoughStrength — Fault Roughness and Strength

The B_RoughStrength GUI is divided into two panels. The left panel controls fault roughness parameters; the right panel controls fault strength (frictional) parameters. Both panels include an Apply to all option, and both operate on a per-segment basis: parameters can be set independently for each fault section in the loaded model.

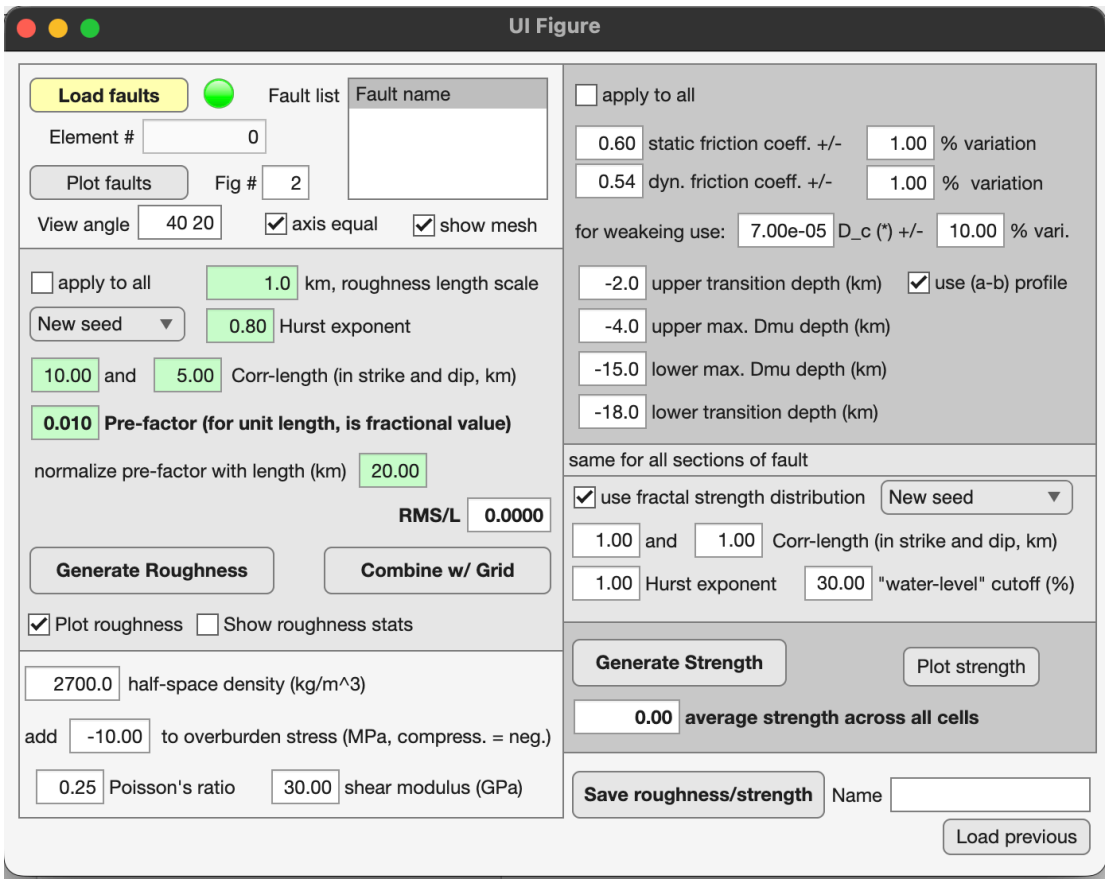

Figure 1: The B_RoughStrength GUI. The left panel handles fault roughness and half-space properties. The right panel handles frictional strength parameters. Load faults at the top left; Save roughness/strength at the bottom right.

Apply to all Both panels include an Apply to all checkbox. When ticked, any value changed in a given field is applied simultaneously to all fault sections, not just the currently selected one. This avoids having to enter identical values individually for each segment in multi-fault models.

Per-segment parameters All roughness and strength parameters are defined independently for each fault section. This allows, for example, one segment to have higher roughness amplitude than another, or different seismogenic depth limits on different faults.

5.1 Loading and Visualising the Fault Model

Press Load faults to open a file browser and select the .ftrg file produced by an A_ GUI. On successful load:

- The green status lamp next to Load faults turns on.
- The Element # field reports the total number of triangular fault elements.
- All fault section names appear in the Fault list box.

Press Plot faults to display the triangulated mesh. The plot controls (Fig #, View angle, axis equal, show mesh) work identically to those in the A_ GUIs. Use this to confirm that the correct model has been loaded before proceeding.

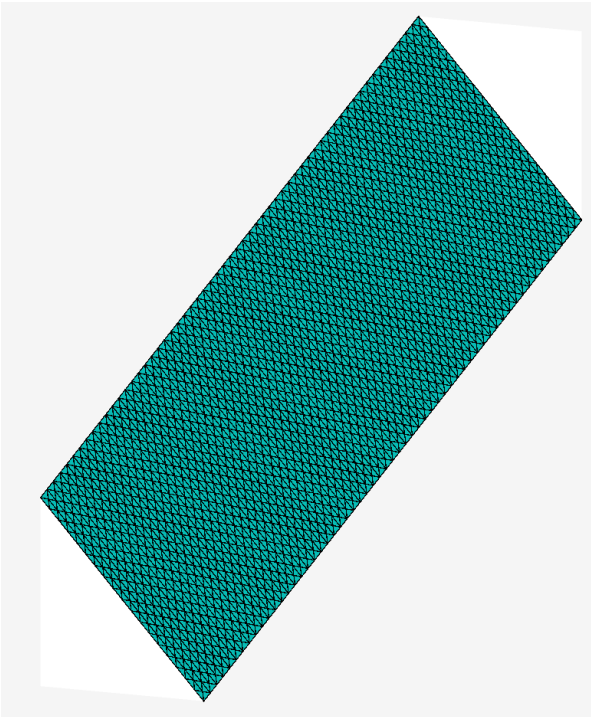

Figure 2: *The fault mesh immediately after loading, before roughness is applied. The uniform teal colouring confirms a flat, unperturbed surface.*

Figure 3: *The same fault after generating and combining roughness (correlation length: 3 km along strike, 5 km along dip; Hurst exponent: 0.80; pre-factor: 0.01 normalised at 20 km). The colour scale reflects out-of-plane topographic displacement.*

5.2 Fault Roughness (Left Panel)

Fault roughness is represented as a two-dimensional random field defined by a von Kármán autocorrelation function. The field describes out-of-plane perturbations of the fault surface and is characterised by three physical parameters plus a length scale.

5.2.1 Roughness Parameters

Parameter (GUI label)	Description
Hurst exponent	Controls how fast roughness amplitude decays toward smaller spatial scales. Lower values produce slower decay (more small-scale roughness); higher values produce smoother surfaces at small scales. Typical range: 0.6–0.9.
Corr-length (in strike and dip, km)	Correlation lengths along strike and dip. These mimic the effect of accumulated slip on fault surface morphology: strike-slip faults typically show longer correlation along strike (groove-like slickensides); dip-slip faults show longer correlation along dip.
Pre-factor (for unit length, is fractional value)	Controls the roughness amplitude at a given reference length scale. Typical values are 0.005–0.01. This is a fractional value: at the normalising length scale, the RMS topography equals pre-factor × that length.

Parameter (GUI label)	Description
normalize pre-factor with length (km)	The reference length scale used to scale/normalize the pre-factor. Using a fixed value here (rather than per-fault length) ensures consistent roughness amplitudes across faults of different sizes in the same model. For example, if a 100 km and a 10 km fault share the same normalising length, their roughness amplitudes are directly comparable.
km, roughness length scale	The spatial resolution at which the roughness field is generated. This should be comparable to the fault mesh element size. For a mesh with ~1 km elements, set this to ~1 km.
RMS/L	Displayed after generating the roughness field. Reports the RMS surface topography normalised by the normalising length — a diagnostic measure of roughness amplitude.

5.2.2 Seed Control

The roughness field is a stochastic realisation: pressing Generate Roughness without conserving the seed will produce a different random field each time. Two options control this behaviour:

Seed option	Behaviour
New seed	A new random seed is drawn each time Generate Roughness is pressed, producing a different realisation.
Previous seed	The same seed as the last generation is reused, reproducing the identical roughness field. Use this to regenerate a specific realization after adjusting other parameters.

5.2.3 Half-Space Properties

The lower portion of the left panel defines the elastic half-space in which the fault is embedded. These values are used to compute overburden (lithostatic) stress, which in turn determines the effective normal stress on the fault.

Parameter (GUI label)	Description
half-space density (kg/m³)	Rock density used to compute lithostatic stress with depth. Default: 2700 kg/m³.
add … to overburden stress (MPa, compress. = neg.)	An additional normal stress correction added to the lithostatic overburden. Negative values increase the effective normal stress. The goal here is to stabilize potentially poorly-behaving fault elements near the surface (where normal stress is low).
Poisson's ratio	Elastic Poisson's ratio of the half-space.
shear modulus (GPa)	Shear modulus of the half-space, used in stress drop and slip calculations.

5.2.4 Roughness Generation Workflow

Follow these three steps to generate and apply roughness:

Step	Action
1	Set all roughness parameters for the selected fault section (or tick Apply to all for uniform values across all sections).
2	Press Generate Roughness to compute the 2D random field at the specified roughness length scale. At this stage the field exists as a regular grid — it has not yet been applied to the triangular mesh.
3	Press Combine w/ Grid to interpolate the roughness field onto the triangular mesh vertices, physically perturbing the fault surface. Inspect the result using Plot roughness.

Important: For dipping faults that reach the surface, adding roughness may push some mesh vertices above the free surface (positive depth). The GUI automatically lowers the entire fault so that its highest point remains just within the half-space. This vertical shift affects near-surface fault behaviour, which is highly sensitive to proximity to the free surface. For shallow faults, consider limiting roughness amplitude or setting the pre-factor to zero.

Important: In complex multi-fault geometries, roughness can cause fault elements to intersect or come into very close proximity with each other. This can produce numerical singularities in the main simulation code. It is the user's responsibility to check that the roughened geometry is free of intersections before saving.

5.3 Fault Strength (Right Panel)

The right panel assigns frictional strength parameters to the fault. Fault strength in MCQsim25 is governed by a slip-weakening friction law: fault elements have a static friction coefficient and a dynamic friction coefficient, and slip is initiated once the static threshold is exceeded. The difference between the two, multiplied by the effective normal stress, determines the available co-seismic stress drop.

5.3.1 Friction Parameters

Parameter (GUI label)	Description
static friction coeff. +/− % variation	Average static friction coefficient across the fault section, with an optional percentage variation (uniformly distributed across elements).
dyn. friction coeff. +/− % variation	Average dynamic friction coefficient. The difference (static − dynamic) determines the available co-seismic stress drop.
D_n (*) +/− % vari.	Slip-weakening distance, defined in relative terms as a multiple of the fault element size rather than in metres. This ensures that single-element failure is physically meaningful regardless of mesh resolution. See Zielke & Mai (2023, BSSA) for the full justification.

5.3.2 Stability Regimes

Not all fault elements are capable of producing unstable (seismogenic) slip. An element's stability is determined by comparing the co-seismic stress drop it can generate against the minimum stress drop required to slip by at least one D_n length. Three stability regimes are possible:

Regime	Behaviour
Unstable	Element has sufficient available stress drop to initiate and sustain rupture. These are the seismogenic elements.
Conditionally stable (Cond. Stable)	The available stress drop is positive but too small for the element to initiate rupture on its own. These elements can be activated by stress transfer from adjacent unstable elements but cannot self-initiate.
Stable	Dynamic friction exceeds static friction at this element (negative friction drop). These elements always resist slip and form aseismic zones. But, if coseismic stress increases sufficiently to exceed the element's dynamic friction strength then these elements can slip coseismically.

Near the surface, lithostatic overburden is low, so the available stress drop is small relative to the element's D_n threshold. This produces a shallow conditionally-stable or stable zone that mimics the velocity-strengthening behaviour observed in the near-surface crust. At depth, rheologic changes due to increasing temperature with depth can similarly create a stable lower boundary to the seismogenic zone.

5.3.3 Depth-Dependent Friction: the (a−b) Profile

Ticking **use (a−b) profile** applies a depth-dependent adjustment to the dynamic friction coefficient, inspired by the rate-and-state friction parameter (a−b) measured in laboratory experiments (measured by e.g., Blanpied et al., 1991; utilized by e.g., Rice et al., 1993). The concept is that the effective friction drop — and thus the seismogenic potential — varies systematically with depth.

Four depth values define a piecewise-linear (a−b) function:

Parameter (GUI label)	Description
upper transition depth (km)	Depth at which (a−b) begins to become negative (velocity-weakening). Above this depth the fault is aseismic (stable).
upper max. Dμ depth (km)	Depth at which the friction drop reaches its maximum positive value in the upper seismogenic zone.
lower max. Dμ depth (km)	Depth at which the friction drop reaches its maximum positive value in the lower seismogenic zone.
lower transition depth (km)	Depth at which (a−b) returns to zero and then becomes positive again. Below this depth the fault is aseismic (stable).

The resulting piecewise-linear profile is automatically scaled so that the average friction drop within the seismogenic zone (between the upper and lower transition depths) equals the prescribed difference between the static and dynamic friction coefficients. This ensures that the user-defined friction values remain physically meaningful regardless of the profile shape.

The (a−b) profile naturally produces aseismic zones at the top and bottom of the fault where the dynamic friction coefficient exceeds the static coefficient (negative friction drop). These zones act as barriers to rupture propagation and mimic the velocity-strengthening behaviour inferred from geodetic and seismological observations.

5.3.4 Fractal Strength Distribution

Ticking **use fractal strength distribution** applies a second, independent random field to spatially redistribute the dynamic friction coefficient across the fault surface. This field is -intentionally- uncorrelated with the geometric roughness field.

The physical motivation is that many factors beyond geometric roughness affect local fault strength: fluid pressure, gauge material distribution, fine-scale lithological heterogeneity, and micro-structural variation. These effects collectively produce fault asperities — regions of elevated strength — that have been inferred from geodetic inversions and laboratory experiments. The fractal strength field provides a bulk proxy for this heterogeneity.

Parameter (GUI label)	Description
Corr-length (in strike and dip, km)	Spatial correlation length of the strength random field. Larger values produce broader, more coherent asperity patches; smaller values produce more diffuse heterogeneity.
Hurst exponent	Decay rate of strength heterogeneity with decreasing spatial scale. Same interpretation as for geometric roughness.
"water-level" cutoff (%)	A percentile threshold applied to the fractal strength field. Elements whose strength falls below this threshold become conditionally stable (their dynamic friction coefficient is raised to equal the static coefficient, eliminating any friction drop). Conceptually, imagine the fractal field as topography: the water level determines how much of the surface is ‘submerged’ and therefore aseismic. Raising the water level isolates stronger asperity patches and reduces the seismogenic area.
Seed (New / Previous)	Controls whether each call to Generate Strength uses a new random seed or reproduces the previous realisation.

Note: *The fractal strength field is currently applied globally (same for all sections of fault), as indicated by the label in the GUI. The correlation length and Hurst exponent for this field are independent of those used for geometric roughness.*

> ***Note:*** *The water-level concept is directly inspired by Lay et al.'s work on subduction zone asperities (e.g., Lay et al., 2012). MCQsim25 applies this principle to all fault types, not limited to subduction zones. The choice of correlation length and water level should reflect the physical process believed to control the observed asperity distribution.*

5.4 Strength Distribution Examples

The Plot strength button displays a four-panel diagnostic figure for the current strength configuration. The panels show: (top left) fault strength (co-seismic stress change in MPa); (top right) D_n distribution; (bottom left) stability regime map; (bottom right) friction coefficients vs. depth colour-coded by stability regime.

The four examples below illustrate how the strength parameters interact.

Example 1 — Uniform Friction, No Depth Variation

Static friction: 0.60; dynamic friction: 0.4; no (a−b) profile; no fractal strength field. The friction drop is constant with depth, so stress drop increases monotonically with overburden. Only the near-surface elements become conditionally stable (low overburden). The fault is almost entirely unstable.

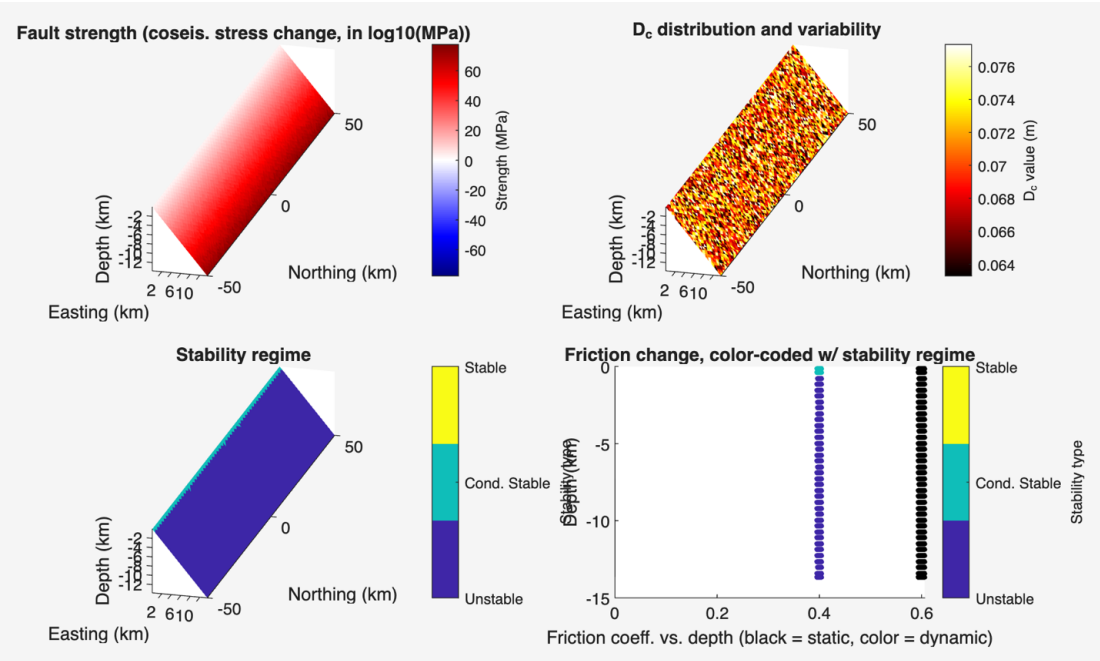

Figure 4: *Uniform friction coefficients (static 0.60, dynamic 0.4) with no depth-dependent profile and no fractal strength field. Fault strength increases smoothly with depth (upper left). The near-surface layer is conditionally stable (cyan) while the remainder of the fault is unstable (blue). Friction coefficients are constant with depth (lower right).*

Example 2 — Depth-Dependent Friction: (a−b) Profile

Same friction coefficients as Example 1, but with the (a−b) profile enabled. The upper and lower aseismic zones are now clearly defined. The friction drop peaks within the seismogenic zone and tapers to zero (or reverses) at the transitions, producing stable zones at both the top and bottom of the fault.

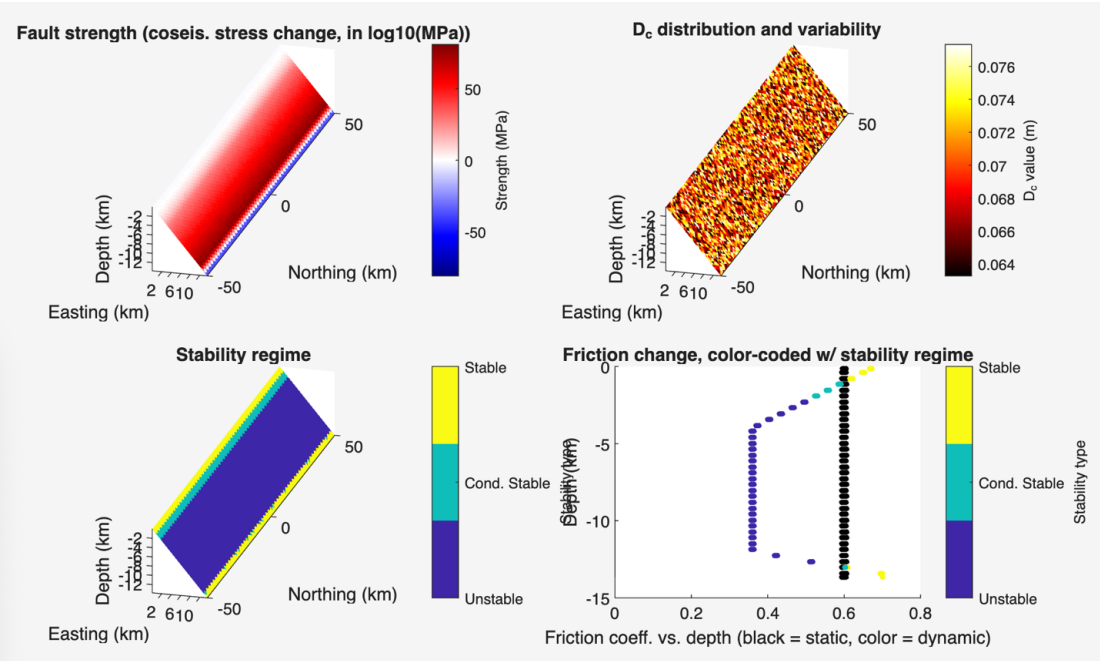

Figure 5: *Depth-dependent friction using the (a−b) profile. Transition depths: upper −1 km, upper max. Dμ −3 km, lower max. Dμ −12 km, lower transition −13 km. Stable aseismic zones (yellow) appear at the top and bottom of the fault, bounding the seismogenic zone. The lower-right panel shows the depth-dependent variation of the dynamic friction coefficient.*

Example 3 — (a−b) Profile with Fractal Strength Distribution

The (a−b) profile from Example 2 is combined with a fractal strength field. The water-level cutoff eliminates the friction drop in the weakest portions of the fault, creating a spatially heterogeneous stability map with discrete asperity patches embedded in a broader conditionally-stable and stable background.

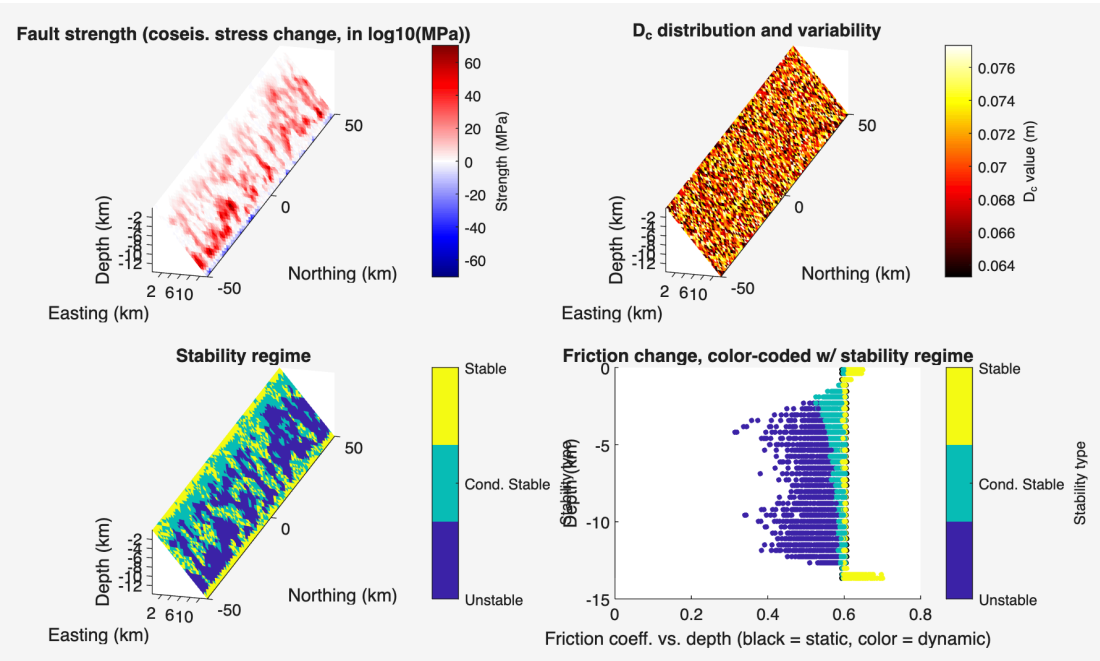

Figure 6: *Combination of the (a−b) profile and a fractal strength distribution. The stability map (lower left) now shows a complex mosaic of unstable (blue) asperities, conditionally stable (cyan) regions, and stable (yellow) patches. The fault strength map (upper left) reflects both the depth trend and the spatial heterogeneity of the fractal field.*

Example 4 — Effect of Increased Correlation Length and Water Level

The same random seed as Example 3 is used (Previous seed), but with a larger correlation length and a higher water-level cutoff. Unstable asperity patches become smaler and more isolated, with a greater fraction of the fault surface rendered conditionally stable or stable. This demonstrates how the water-level and correlation length together control the size and spatial organisation of seismogenic asperities.

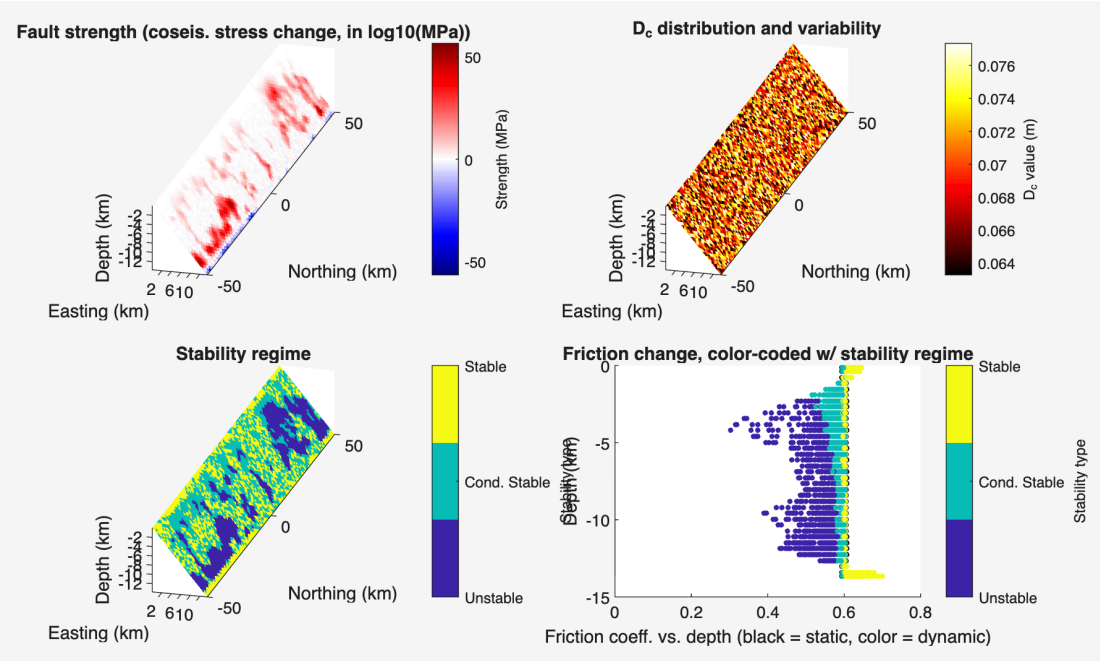

Figure 7: *Same seed and (a−b) profile as Example 3, but with a larger fractal strength correlation length and a higher water-level cutoff. Asperity patches are more broadly connected and the stable (yellow) fraction of the fault has increased, illustrating the sensitivity of the seismogenic pattern to these parameters.*

5.5 Step-by-Step Workflow

The complete B_RoughStrength workflow, from loading to saving, follows these steps:

Step	Action
1	Press Load faults and select the .ftrg file. Confirm the correct model using Plot faults.
2	Select a fault section from the Fault list (or tick Apply to all). Set roughness parameters: Hurst exponent, correlation lengths, pre-factor, normalising length, and roughness length scale. Choose seed behaviour (New / Previous). Do this for all fault sections.
3	Press Generate Roughness to create the 2D random field. Review RMS/L. To skip roughness, set the pre-factor to zero but still press this button.
4	Press Combine w/ Grid to apply the roughness field to the triangular mesh. Inspect visually with Plot roughness. (3) and (4) will perform their task for all fault sections.
5	Set half-space properties: density, overburden correction, Poisson's ratio, and shear modulus.
6	Set friction parameters: static and dynamic friction coefficients, D_n value, and optional % variations. Enable use (a−b) profile if desired and set the four transition depths.
7	Optionally enable use fractal strength distribution and set its correlation lengths, Hurst exponent, and water-level cutoff. Choose seed behaviour.
8	Press Generate Strength. Inspect using Plot strength and check the stability map and friction vs. depth panel.
9	Press Save roughness/strength. The GUI writes a .rgh file and a .stg file into the same directory as the .ftrg file. These two files, together with the geometry from the A_ GUI, are all that is required to run the main simulation.

Important: Careful parameter space analysis is required before running the main simulation. The strength configuration strongly influences the resulting seismicity patterns, earthquake magnitudes, recurrence behaviour, as wells as scaling relationships. It is strongly recommended to inspect the Plot strength output for every new parameter combination before saving and to adjust them if necessary after catalog inspection.

6 Next Steps

After saving from B_RoughStrength, the full set of input files required by the main simulation code is complete:

File	Source	Content
.rgh	B_RoughStrength	Fault roughness field interpolated onto the triangular mesh.
.stg	B_RoughStrength	Fault strength (friction) parameters for each element.

> ***Note:*** *The .ftrg file from the A_ GUI is not used by the main simulation code. Only the .rgh and .stg files are required as direct inputs. The .rgh file "inhereted" the fault element's tectonic loading information from .ftrg.*

The workflow continues:

Step	Description
Main Code	Run the stochastic earthquake simulation using the .rgh and .stg files.
Output GUIs	Visualise and analyse simulation results.

> ***Note:*** *Detailed documentation for the main simulation code and the output GUIs will be provided in subsequent sections of this document.*

MCQsim25

Software Documentation

Main Simulation Code

Overview

The MCQsim25 simulation engine is a terminal-based MPI program. It reads the fault geometry and strength files produced by the input GUIs, runs a stochastic earthquake sequence for a specified catalogue length, and writes earthquake catalogues and quasi-dynamic rupture models to binary output files. These outputs are then visualised using the MATLAB-based output GUIs.

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7 Running MCQsim25

This section covers the practical steps required to compile, configure, and run the simulation: which input files are needed, how to set the parameter file, how to launch the executable, and what output files are produced.

7.1 Input Files

All input files that are used in simulation must reside in the same directory as the MCQsim25 executable. They share a common **root name** (referred to as `InputName` in the parameter file). For example, if the root name is `FaultModel`, the files are named as shown in the table below.

File name	Format	Content
FaultModel.txt	ASCII	Parameter file. Controls all simulation settings (catalogue length, seed, output switches, etc.). Must be present and correctly formatted.
FaultModel.ftxt	ASCII	Geometry summary file. Human-readable description of the fault geometry. Not used by the simulation engine.
FaultModel.ftrg	Binary	Original fault geometry file (without roughness). Not used by the simulation engine; superseded by the .rgh file.
FaultModel_Summary_RoughStrength.txt	ASCII	Human-readable summary of the roughness and strength configuration applied in B_RoughStrength. Not used by the simulation engine.
FaultModel_1.rgh	Binary	Fault geometry file including roughness. Equivalent to .ftrg but with the roughened surface. Required by the simulation.
FaultModel_1.stg	Binary	Fault element friction and strength parameters. Required by the simulation.

> ***Note:*** *Only the .rgh and .stg files are read by the simulation engine. The .ftrg file, the .ftxt summary, and the RoughStrength summary text are for reference only.*

7.2 Parameter File

The parameter file (`FaultModel.txt`) controls the simulation run. Its structure is fixed: do not add or remove lines, and do not insert spaces in the parameter labels. The file must begin with a version number.

7.2.1 Example Parameter File

A complete example parameter file is shown below:

```
VersionNumber                    202511
Run_Parameter_Information
-------------------------
```

```
InputName:                      FaultModel
PlotCatalog2Screen(0/1):        1
RandSeedValue:                  1001
MinElementNum4Catalog:          1
--------------------------
StoreSTF4LargeEQs(0/1):         1
MinMagnitude4STF:               7.0f
--------------------------
Visco_AfterSlip(years):         3.0f
--------------------------
HealingDuringCoseisFraction:    0.05f
OvershootFraction:              1.2f
--------------------------
EQrecordLength(years):          5000.0f
```

7.2.2 Parameter Reference

Parameter	Description
`InputName:`	Root name shared by all input files. For example, FaultModel matches FaultModel.rgh, FaultModel.stg, etc.
`PlotCatalog2Screen(0/1):`	Print catalogue information to the terminal during the simulation (1 = yes, 0 = no). Typically left at 1 to monitor progress.
`RandSeedValue:`	Base seed for the random number generator. Some parameters (e.g., post-seismic friction recovery) can vary stochastically during the simulation; the seed controls this variability. Each MPI process receives a unique seed derived by adding its MPI rank to this base value. To reproduce an identical catalogue exactly, use the same seed and the same number of MPI processes.
`MinElementNum4Catalog:`	Minimum number of activated fault elements for an event to be written to the catalogue. Set to 1 to record all events. Set to a higher value to suppress recording of very small events. Small events still occur and influence stress evolution — they are simply not written to file.
`StoreSTF4LargeEQs(0/1):`	Save the quasi-dynamic rupture evolution (source time function, STF) of individual large events to .srfb files (1 = yes, 0 = no).
`MinMagnitude4STF:`	Minimum moment magnitude an event must reach for its quasi-dynamic rupture model to be saved. Example: 7.0f saves all M ≥ 7.0 events.
`Visco_AfterSlip(years):`	Characteristic relaxation time (years) for postseismic after-slip along the fault, modelled with a Maxwell spring-dashpot. This is the time for the postseismic signal to decay to 1/e (~35%) of its initial value. A value of 3.0 means the after-slip contribution decreases by ~65% within the first three years. Higher values correspond to higher effective viscosity.
`HealingDuringCoseisFraction:`	Fractional increase in friction coefficient applied per coseismic iteration step to elements that have stopped slipping while the rupture front is still propagating. Controls the rate at which a healed element regains strength and resists re-rupture during the same event. See Section 8.4 for details.

Parameter	Description
`OvershootFraction:`	Ratio of dynamic arrest friction to the pre-event static friction coefficient. Values greater than 1.0 produce overshoot (the fault overshoots its equilibrium slip before arresting), mimicking dynamic undershoot/overshoot behaviour observed in kinematic rupture models.
`EQrecordLength(years):`	Total catalogue length in years. If an existing catalogue is extended (continued run), this value specifies the additional years to simulate, not the total.

Important: The parameter file structure is fixed. Do not add, remove, or reorder lines. Do not insert spaces within parameter label strings (e.g., InputName: is correct; Input Name: will cause a read error).

7.3 Launching the Simulation

Once all input files are in place, start the simulation from the terminal using mpirun:

```
mpirun -n 6 ./MCQsim25 FaultModel.txt
```

This command launches MCQsim25 as an MPI job using **6 CPU processes**. The parameter file `FaultModel.txt` is passed as the sole command-line argument. The number of processes can be adjusted; see the note on seed reproducibility below.

Important: To reproduce an identical earthquake catalogue, you must use the same RandSeedValue and the same number of MPI processes (-n). Each process derives its individual seed by adding its MPI rank to the base seed, so changing the number of processes changes the per-process seeds and therefore the resulting catalogue.

7.4 Output Files

MCQsim25 writes several output files on completion. They share the same root name as the input files and are placed in the same directory. The output GUIs (Section 9) are used to open and visualise these files.

File name	Format	Content
FaultModel.pre	Binary	Pre-run snapshot. Records friction properties, fault strength, and long-term slip rate of each fault element at simulation start. Useful for verifying initial conditions and inspecting the pre-seismic state.
FaultModel.cat	Binary	Earthquake catalogue. Contains the full record of simulated events meeting the MinElementNum4Catalog threshold: event time, magnitude, hypocenter, and participating elements.
FaultModel_M7.38_t3845.8.srfb	Binary	Quasi-dynamic rupture model for a single large event. The filename encodes the event magnitude and origin time. This is a binary version of the standard rupture format (*.srf, Graves 2002). One file is written per event exceeding MinMagnitude4STF.

Catalogue visualisation: Open `.cat` files with GUIs named starting with `Z_` (e.g., `Z_Catalog_MCQsim_TREAD`, `Z_MCQsim_LoadCatalog`). These GUIs also allow export to `.mat` or `.txt` format.

Rupture model visualisation: Open `.pre` and `.srfb` files with GUIs named starting with `Y_` (e.g., `Y_PreRunAndSRF_MCQsim_Tread`).

8 Simulation Physics

This section describes how the main simulation code implements each phase of the earthquake cycle. The descriptions reflect the actual numerical approach used internally, and are intended to help users understand how their parameter choices influence simulation behaviour.

8.1 Tectonic Loading

MCQsim25 accepts two types of tectonic loading, both specified in the B_RoughStrength GUI: a prescribed fault slip rate or a regional stress rate (see Section 5). Internally, both are converted into a per-element stressing rate using the same redistribution procedure, but with an important difference in how the result is scaled.

8.1.1 Loading from a Prescribed Fault Slip Rate

A direct back-slip approach would concentrate stress at fault edges, producing artefactual stress concentrations at the boundary of the fault model. MCQsim25 avoids this by redistributing the slip rate before computing stresses, using the following procedure:

Step	Description
1	Each element's prescribed slip rate is converted to a self-stress using that element's own stiffness (self-stiffness). At this stage there is no interaction between elements.
2	The resulting stress field is iteratively relaxed: excess stress above the element strength is released, and during this process each element's slip is recorded. Elements communicate with each other in this step through elastic interaction. Approximately 200 iterations are sufficient for convergence.
3	The recorded slip amounts (from Step 2) define a new, redistributed slip-rate field across the fault. This field is scaled to match the prescribed input slip rate, preserving total seismic moment. As a direct consequence, slip rates taper naturally toward fault edges without any artificial boundary treatment.
4	The redistributed slip-rate field is applied using the standard back-slip method to compute the long-term stressing rate for each element.

8.1.2 Loading from a Regional Stress Rate

This follows the same redistribution procedure as Section 8.1.1, but with the regional stress field (resolved onto the fault plane) substituted for the initial self-stress in Step 1. One important difference applies in Step 3:

Because no input slip rate is prescribed, the redistributed slip-rate field **cannot be scaled**. The absolute stressing rate therefore depends on element size: from Hooke's law, the same stress produces larger slip in larger ("softer") elements and smaller slip in smaller ("stiffer") elements. When using regional stress loading, the stress magnitude must be chosen and tuned so that the resulting long-term slip rates are consistent with observational constraints (e.g., geodetic or geologic slip-rate estimates).

> ***Note:*** *The element-size dependence of regional stress loading means that mixed-resolution meshes may produce inconsistent slip rates across sections. Uniform mesh resolution is recommended when using this loading mode.*

8.2 Interseismic Behaviour

Following the tectonic loading calculation, each fault element has a nominal stressing rate (change in along-strike and along-dip stress per year). During the interseismic phase, elements accumulate stress at this rate. The response when an element's stress reaches its current strength depends on its stability regime:

Stability regime	Behaviour when strength is exceeded
Unstable	Signals the onset of an earthquake. The simulation switches immediately to the coseismic phase.
Conditionally stable / Stable	Cannot self-initiate rupture. Excess stress above the current strength is redistributed interseismically to neighbouring elements via elastic interaction. This corresponds physically to aseismic creep, though the actual slip magnitude depends on proximity to locked (unstable) asperities. In effect, conditionally stable and stable elements act as stress conduits, loading adjacent asperities.

8.3 Coseismic Phase

8.3.1 Rupture Initiation

As soon as an unstable element's stress exceeds its static strength, the coseismic phase begins. At this moment, all conditionally stable and stable elements undergo an immediate strength increase: their frictional resistance is raised from the static to the dynamic level. This stiffening makes them harder to rupture and confines the initial propagating front to the unstable elements unless sufficient stress transfer forces them to participate.

Subsequently, any element whose applied stress exceeds its (now dynamic) strength joins the rupturing event. The rupture propagates by elastic stress transfer from slipping elements to their neighbours.

8.3.2 Coseismic Time

The underlying stress transfer expressions are quasi-static. To introduce a physical timescale, MCQsim25 uses the P-wave velocity V_p (derived from the half-space elastic properties) to assign a travel time to each coseismic iteration step. Specifically, the code uses edge dislocations and computes the travel time for an elastic signal to propagate from a triangle's edge to the in-centre of the receiving element, taking the average edge-to-centre distance as the characteristic length. This approach was inspired by Ward (2002, 2012).

8.3.3 Healing During Rupture

Elements that stop slipping during an ongoing rupture (because local stress has dropped below their dynamic strength) can partially regain strength coseismically. This is controlled by `HealingDuringCoseisFraction` (f). At each subsequent coseismic iteration step, the gap between the element's current strength and its pre-rupture static strength is reduced multiplicatively:

```
gap(t+1) = gap(t) × (1 − f)
```

For example, with $f = 0.05$: an element whose strength gap is 1.0 at the moment it stops slipping will have a gap of 0.95 after one iteration step, 0.90 after two, 0.86 after three, and so on. As strength recovers, re-rupture of the healed element during the same event becomes progressively less likely.

8.4 Postseismic Behaviour

After the coseismic phase ends, elements begin recovering their pre-rupture static strength. The process differs by stability regime:

Stability regime	Postseismic behaviour
Unstable	Recovers to static strength generally immediately after the coseismic phase concludes. Resumes normal interseismic stress accumulation.
Conditionally stable / Stable	May have accumulated co-seismically transferred stress that exceeds their static strength threshold. Because they cannot rupture seismically, this excess is released over a relaxation timescale rather than instantaneously.

Stability regime	Postseismic behaviour
	During each subsequent interseismic loading step, these elements first receive their nominal stressing rate, compare available stress against current strength, and release any excess through quasi-static redistribution. Their frictional strength decays back toward the static level according to the Maxwell relaxation time defined by Visco_AfterSlip(years).

The postseismic after-slip model follows a Maxwell spring-dashpot rheology along the fault. The parameter `Visco_AfterSlip(years)` defines the time constant τ at which postseismic slip decays to $1/e$ (≈35%) of its initial amplitude. A value of 3.0 years means ~65% of the postseismic signal is released within the first three years; a higher value corresponds to a more viscous (sluggish) response.

8.5 Earthquake Cycle Summary

The four phases described above repeat continuously throughout the simulation. The following table summarises how the key parameter file entries map onto the physical processes:

Phase	Parameter(s)	Physical role
Tectonic loading	`InputName`	Selects the .rgh / .stg files that define the slip-rate or stress-rate loading.
Interseismic	`EQrecordLength`	Total simulation duration; controls how many seismic cycles are simulated.
Coseismic — time	Half-space properties (from B_GUI)	V_p controls the coseismic iteration time step.
Coseismic — healing	`HealingDuringCoseisFraction`	Rate at which healed elements resist re-rupture during an ongoing event.
Coseismic — arrest	`OvershootFraction`	Dynamic arrest friction relative to static friction; controls slip overshoot.
Postseismic	`Visco_AfterSlip(years)`	Maxwell relaxation time for after-slip on conditionally stable and stable elements.
Catalogue	`MinElementNum4Catalog` / `MinMagnitude4STF`	Thresholds for writing events and rupture models to file.
Reproducibility	`RandSeedValue`	Base seed; must be combined with the same -n value to reproduce an identical catalogue.

9 Next Steps

Once the simulation has completed, the output files are ready for visualisation and analysis using the MATLAB-based output GUIs:

GUI prefix	Opens	Purpose
`Z_`	`.cat`	Earthquake catalogue: magnitude–frequency distributions, spatial maps, recurrence analysis, export to .mat or .txt.
`Y_`	`.pre` / `.srfb`	Pre-run state and quasi-dynamic rupture models: slip distributions, rupture velocity, source time functions.

Note: *Detailed documentation for the output GUIs will be provided in subsequent sections of this document.*

MCQsim25

Software Documentation

Output Component: Visualisation GUIs

Overview

MCQsim25 produces two classes of output file: a pre-run snapshot and an earthquake catalogue, together with quasi-dynamic rupture models for selected large events. Two families of MATLAB GUIs are provided to visualise these outputs. GUIs prefixed with Y_ handle the pre-run state and individual rupture models; GUIs prefixed with Z_ handle the earthquake catalogue.

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10 Output GUIs

The two output GUI families are summarised below. Both follow the same numbered-button convention used throughout the MCQsim25 workflow.

GUI prefix	Opens	Purpose
`Y_`	`.pre` / `.srfb`	Pre-run state visualisation and quasi-dynamic rupture animation.
`Z_`	`.cat`	Earthquake catalogue visualisation, spatial mapping, and export.

10.1 Y_ — Pre-Run and Rupture Visualisation

The Y_ GUI (Figure 1) loads the `.pre` file produced by the simulation and, optionally, individual `.srfb` quasi-dynamic rupture model files. It is divided into three functional areas: the left panel for pre-run data, the centre panel for plot options, and the right panel for rupture model playback.

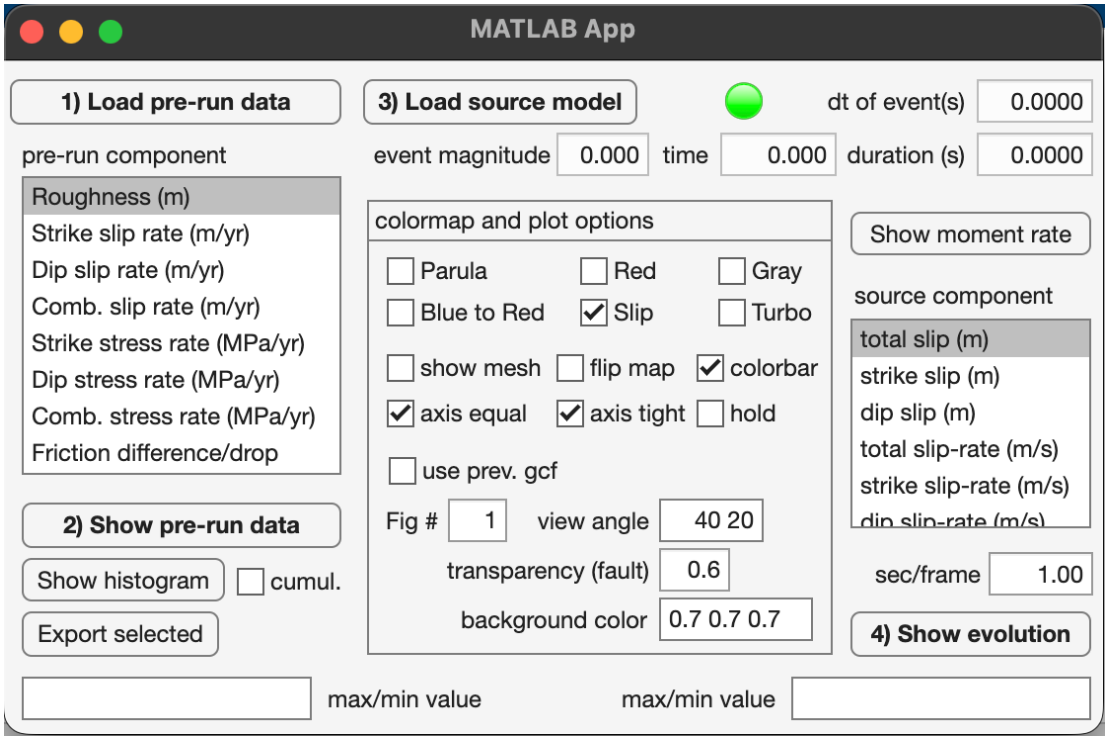

Figure 1: *The Y_ GUI. Left panel: pre-run component list and visualisation controls. Centre panel: colormap and plot options shared by both pre-run and rupture views. Right panel: rupture model loading, source time function display, and animation controls.*

10.1.1 Loading Data

Button / action	Description
1) Load pre-run data	Loads the .pre file. This also imports the fault geometry, which is required before any .srfb rupture file can be visualised. Always load the pre-run file first.

Button / action	Description
3) Load source model	Loads a .srfb quasi-dynamic rupture file for a single event. The event magnitude, origin time, duration, and coseismic time step (dt) are displayed automatically. Requires the .pre file to be loaded first.

> ***Note:*** *The .srfb files do not contain fault geometry. The .pre file must be loaded first to provide the geometric context for rupture visualisation.*

10.1.2 Pre-Run Data Inspection

After loading, the left-panel list box shows all available pre-run components (Roughness, Strike/Dip/Combined slip rate, Strike/Dip/Combined stress rate, Friction difference/drop). Selecting an entry updates the **max/min value** fields at the bottom of the panel automatically. Adjust these values to set the colour range before pressing 2) Show pre-run data to plot.

10.1.3 Plot Options

The centre panel provides shared plot controls for both pre-run and rupture views:

- **Colormap:** Parula, Red, Gray, Blue to Red, Slip, Turbo.
- **Display toggles:** show mesh, flip map, colorbar, axis equal, axis tight, hold.
- **use prev. gcf:** plots into the currently active MATLAB figure rather than opening a new one.
- **Fig #, view angle, transparency, background color:** standard plot formatting controls.

10.1.4 Rupture Model Playback

Once a .srfb file is loaded, the right panel becomes active:

- **source component list box:** selects the quantity to display — total slip (m), strike slip (m), dip slip (m), total slip-rate (m/s), strike slip-rate (m/s), or dip slip-rate (m/s).
- **Show moment rate:** plots the event's moment rate function (source time function).
- **4) Show evolution:** plays a frame-by-frame animation of the selected source component evolving across the fault surface. Frame rate is controlled by sec/frame.
- **max/min value (right):** sets the colour range for the rupture animation independently from the pre-run colour range.

10.2 Z_ — Earthquake Catalogue Visualisation

The Z_ GUI (Figure 2) loads the `.cat` catalogue file produced by the simulation and provides tools for plotting, spatial mapping, temporal analysis, and export.

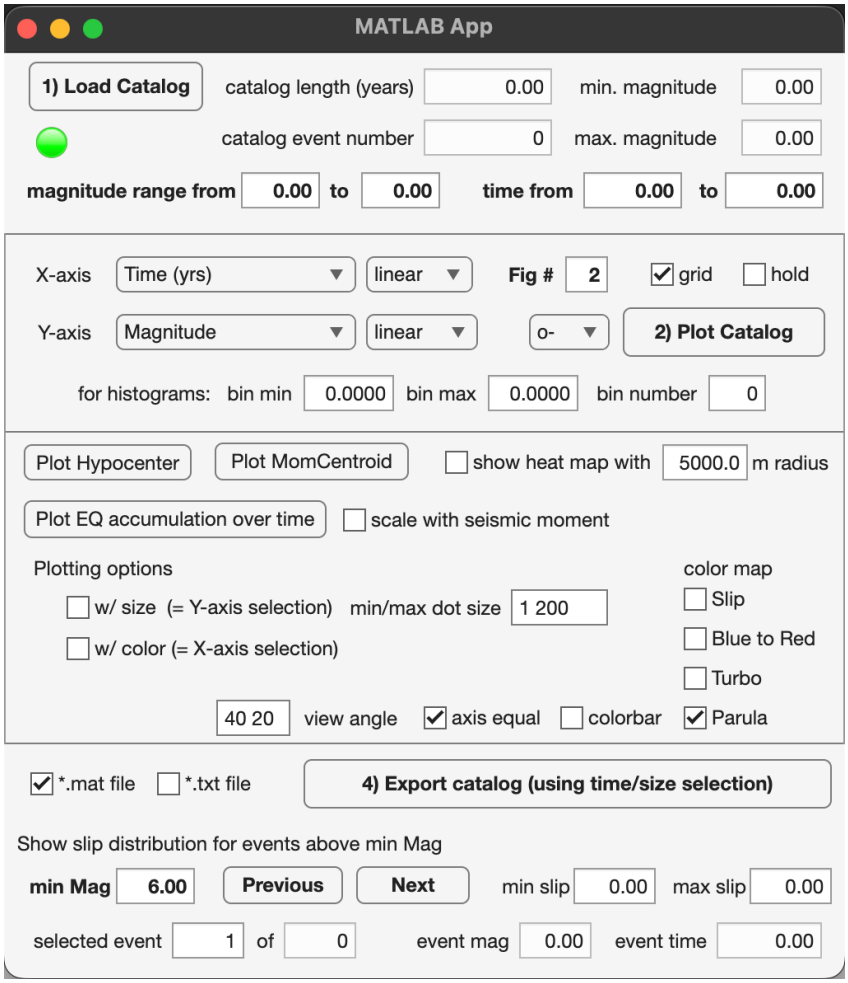

Figure 2: *The Z_ Catalogue GUI. Load controls at the top; axis and histogram settings in the centre panel; spatial and temporal plot buttons below; export and slip distribution tools at the bottom.*

10.2.1 Catalogue Loading and Cropping

Press **1) Load Catalog** to open the `.cat` file. The fields at the top of the GUI report the full catalogue statistics (length in years, total event count, min/max magnitude) — these are read-only and reflect the complete record as written by the simulation.

The **bold** row of fields immediately below (**magnitude range from … to …** and **time from … to …**) defines the active subset used for all subsequent plotting and export operations. Use these to exclude the simulation spin-up phase or to focus on a specific magnitude range without reloading.

10.2.2 Catalogue Plotting

The centre panel controls the main scatter plot:

- **X-axis / Y-axis dropdowns:** select any catalogue attribute (Time, Magnitude, Depth, etc.) for each axis.
- **Scale:** linear or log per axis.
- **Symbol style:** set via the drop-down (e.g., o−).
- **w/ size:** scales symbol size by the Y-axis selection (min/max dot size sets the range).
- **w/ color:** colours symbols by the X-axis selection using the chosen colormap.
- **grid / hold:** standard MATLAB plot options.

10.2.3 Spatial Mapping

Plot Hypocenter and **Plot MomCentroid** display event locations on the fault surface. Enabling **show heat map** with a specified radius (metres) identifies spatial clusters of epicentres. Note that the heat map computation involves pairwise distance checks and is slower than simple scatter plots.

10.2.4 Temporal Analysis

The **Plot EQ accumulation over time** button produces a cumulative event count (or moment release) plot over simulation time — colloquially called the 'hockey-stick' plot because of the characteristic shape that emerges during activity bursts. Ticking **scale with seismic moment** replaces event counts with cumulative seismic moment, highlighting variations in fault productivity.

10.2.5 Slip Distribution of Individual Events

The bottom section of the GUI allows browsing through the slip distribution of individual events. Set **min Mag** to the desired threshold and then use **Previous** / **Next** to step through qualifying events. The selected event number, magnitude, and origin time are displayed, and the slip distribution is plotted on the fault surface (total combined slip only).

> **Important:** The min Mag threshold for slip distribution browsing must be set before loading the catalogue (before pressing 1) Load Catalog). To use a different threshold, update the value and reload the catalogue.

10.2.6 Export

Press **4) Export catalog** to save the currently cropped catalogue (respecting the active magnitude range and time window) as a `.mat` or `.txt` file, selectable via the checkboxes at the bottom left.

11 Complete MCQsim25 Workflow

The table below summarises the end-to-end MCQsim25 workflow across all components documented in this manual.

#	Component	Tool	Output / result
1	Fault geometry	A_GEOMfromSTL or A_GEOMmanual	Triangulated fault mesh (.ftrg). Fault sections with names, IDs, and loading assignments.
2	Roughness & strength	B_RoughStrength	Fault roughness and strength files (.rgh, .stg) ready for simulation.
3	Simulation	MCQsim25 (terminal, MPI)	Earthquake catalogue (.cat), pre-run snapshot (.pre), and rupture models (.srfb) for large events.
4	Rupture inspection	Y_ GUI	Visual inspection of pre-run state (slip rates, stress rates, roughness) and animation of individual earthquake ruptures.
5	Catalogue analysis	Z_ GUI	Catalogue plotting, spatial and temporal analysis, slip distribution browsing, and export.

Note: *This concludes the MCQsim25 software documentation. For further details on specific parameters, scaling relationships, or the underlying physics, refer to Zielke & Mai (2023, BSSA) and the references cited therein.*